

MIX GRADE	DESIGN REF. CODE	REVISION	STATUS
M25	OR20F207	0	APPROVED
MIX DESCRIPTION	CREATED	APPROVED	
General Purpose Concrete — Standard Grade	20 Jul 2026 Pallavi Nandakumar	21 Jul 2026 Quality Control Engineer	

MIX DESIGN SUMMARY		
Compressive Strength (f'ck) at 28 days 25.0 MPa	Standard Deviation (SD) 4.0 MPa	Maximum Aggregate Size 20 mm
Target Mean Strength (f'ck + 1.65 x SD) 31.6 MPa	Free Water / Binder Ratio (W/B) 0.450	Water Content 158 kg/m3
Total Binder (Cement + Fly Ash) 351 kg/m3	Target Workability at Site 100-125 mm (Slump)	Design Density 2,480 kg/m3

MIX PROPORTIONS (SSD/m3)	
Cement	320 kg/m3
Fly Ash (PFA) — 9% of Binder	31 kg/m3
Total Binder (Cement + Fly Ash)	351 kg/m3
Free Water Content	158 kg/m3
Free Water / Binder Ratio (W/B)	0.450
Fine Aggregate	720 kg/m3
Coarse Aggregate — 20 mm	650 kg/m3
Coarse Aggregate — 12.5 mm	460 kg/m3
Total Aggregate	1,830 kg/m3
Admixture — Sika ViscoCrete 3000, 0.60% of Binder	2.10 kg/m3
Design Density	2,480 kg/m3

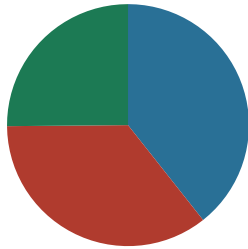
MATERIALS — TYPE / SOURCE / SP. GR.		
MATERIAL	TYPE / SOURCE	SP.GR.
Cement	OPC 53 Grade / Dalmia Cement / Ramco Cements Ltd (approved vendor list)	3.15
Fly Ash (PFA)	Class F Fly Ash / NTPC Kayamkulam Thermal Plant	2.20
Fine Aggregate	M-Sand / Local Quarry, Kasaragod	2.65
Coarse Aggregate — 20 mm	Crushed Granite 20mm / Local Quarry, Kasaragod	2.70
Coarse Aggregate — 12.5 mm	Crushed Granite 12.5mm / Local Quarry, Kasaragod	2.70
Admixture	Sika ViscoCrete 3000	1.09
Maximum Aggregate Size	20 mm	—

AGGREGATE PROPORTIONS		
FRACTION	QTY (kg/m3)	% OF TOTAL
Fine Aggregate	720	39%
Coarse — 20 mm	650	36%
Coarse — 12.5 mm	460	25%
Total	1,830	100%

Coarse / Fine Aggregate Ratio



Aggregate Proportion



- M-Sand (fine) — 39%
- 20 mm (coarse) — 36%
- 12.5 mm (coarse) — 25%

NOTE
1. Admixture dosage to be adjusted based on ambient temperature and transport time.
2. Moisture correction of aggregates to be carried out daily.
3. All quantities are in SSD condition unless specified.

Quality Control Engineer
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Disclaimer: This mix design has been prepared in accordance with applicable Indian Standards based on laboratory trials and the materials specified herein. Actual field performance is dependent upon proper batching, transportation, placement, compaction, finishing, curing and site conditions. Any change in material source or characteristics may require review and revision of the mix design.